RieszNet and ForestRiesz: Automatic Debiased Machine Learning with Neural Nets and Random Forests

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Background: Causal Machine Learning Is Transforming Decision-Making

- We see data-driven causal inference guiding healthcare, public policy, and online platforms.
- We need not just predictions but credible effect estimates with uncertainty.
- We rely on reliable confidence intervals to enable safe policy deployment and auditing.

Background: Many Targets Are Linear Functionals of Regressions

- We focus on targets that reduce to linear functionals of the regression function g0(Z), the conditional mean of Y given Z.
- We study examples such as average treatment effects, off-policy value, average derivatives, and incremental policy effects.
- We use the linear functional view to unify diverse causal and policy estimands.

The Problem: Achieving Root-n-Accurate Debiased Estimation in Practice

- We find plug-in estimators inherit regularization bias from our regression learner.
- We observe instability when propensities are near zero or one and in density tails.
- We note that manual derivations per functional are cumbersome and error-prone.

Opportunity: Riesz Representation Enables Stable Debiasing

- We leverage the Riesz representation, which links linear functionals to an inner product with a representer alpha0.
- We construct a doubly robust estimator by adding a learned correction alpha(Z) times the residual Y minus g(Z).
- We obtain root-n accuracy when the product of the errors in alpha and g is small.

Our Core Idea & Contributions: Automatic Riesz Learning (Auto-DML)

- We learn the Riesz representer alpha0 directly from data using a targeted loss.
- We introduce RieszNet (a multitask neural network) and ForestRiesz (a generalized random forest approach).
- We require only black-box access to evaluate the functional, avoiding analytic derivations.

Method Overview: How We Learn the Riesz Representer

- We jointly learn g and alpha with shared features using a Riesz-targeting loss that uniquely identifies the representer via black-box evaluations of the functional.
- We train end-to-end and cross-fit to construct a doubly robust estimate with valid inference.
- We avoid unstable inverse propensity and density models by learning alpha directly.

RieszNet: Multitask Architecture

- We use a shared representation of Z that feeds two heads: one for alpha and one for g.
- We add a targeted TMLE-style linear augmentation that nudges the regression toward reducing bias identified by alpha.
- We align features end-to-end with the signal in the Riesz representer to improve stability.

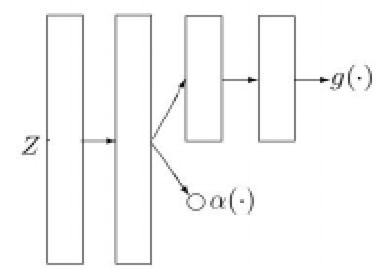

Figure: RieszNet architecture.

RieszNet Objective and Training

- We optimize a regression loss plus a Riesz-targeting loss and a TMLE-targeting loss, together with standard L2 regularization.
- We keep the targeting coefficient unpenalized to enable focused bias reduction.
- We train with stochastic gradient methods, early stopping, and simple cross-fitting.

ForestRiesz: Locally Linear Riesz via GRF

- We model alpha as locally linear in a smooth basis over treatment and covariates.
- We solve local moment equations using generalized random forests.
- We use a split criterion that penalizes ill-conditioned local Jacobians for robustness.

ForestRiesz: Multitasking and Cross-Fitting

- We jointly learn equations for alpha and g within one forest with simple cross-fitting.
- We acknowledge that double cross-fitting can preclude multitasking, creating trade-offs.
- We find that simple cross-fitting with multitasking performs best overall in practice.

DR Estimation, Rate, and Inference

- DR estimator: $\hat{\theta} = \mathbb{E}_n[m(W; \hat{g}) + \hat{\alpha}(Z)(Y - \hat{g}(Z))]$.
- We obtain root-n normality when the errors in alpha and g shrink fast enough.
- We estimate variance from the influence function and construct Wald confidence intervals with near-nominal coverage.

Experimental Setup: Datasets and Targets

- We evaluate on IHDP (semi-synthetic): average treatment effect, sample size 747, 25 covariates.
- We analyze BHP gasoline demand (semi-synthetic): average derivative, sample size 3,640.
- We stress-test across linear and nonlinear confounding scenarios with varying complexity.

Experimental Setup: Baselines and Protocols

- We compare against Dragonnet, random forest plug-in, and CausalForest baselines.
- We report Direct, IPS, and DR (Auto-DML) estimators.
- We use cross-fitting for nuisance learners and evaluate post-TMLE where relevant.

IHDP ATE: Main Accuracy Results

- We obtain MAE 0.110 $\pm$ 0.003 with DR using RieszNet, and 0.126 $\pm$ 0.004 with DR using ForestRiesz.
- We observe Dragonnet at 0.146 $\pm$ 0.010; random forest plug-in at 0.389; CausalForest at 0.728.
- We find DR consistently more stable than Direct or IPS across resamples.

IHDP ATE: Coverage and Distribution

- We achieve near-nominal 95% coverage, typically around 95 to 96 percent.
- We see approximately normal estimate distributions with small bias.
- We find Direct undercovers due to underestimated variance.
- Left: bias, RMSE, and coverage across runs. Right: the sampling distribution of DR estimates is approximately normal.

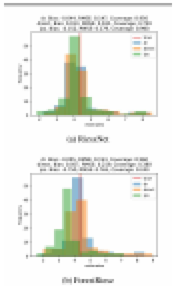

Figure: Bias, RMSE, coverage, and estimate distributions for RieszNet and ForestRiesz on IHDP semi-synthetic datasets (resampled treatment).

BHP Gasoline Demand: Average Derivative Results

- We find ForestRiesz with DR plus post-TMLE achieves bias -0.082, RMSE 0.327, coverage 0.953.
- We observe ForestRiesz with DR achieves bias -0.131, RMSE 0.377, coverage 0.909.
- We report RieszNet with DR at bias 0.062, RMSE 0.504, coverage 0.877.
- We note the random forest plug-in baseline at RMSE 0.327 and coverage 0.912; ForestRiesz slightly improves coverage.

BHP: Distributional Behavior and Coverage

- We see DR estimates centered near the truth with near-normal sampling behavior.
- We find post-TMLE improves bias and coverage for ForestRiesz.
- We observe Direct undercoverage despite low bias due to underestimated variance.
- Left: bias, RMSE, and coverage across 1000 datasets. Right: the DR estimate distribution centers near the truth and is close to normal.

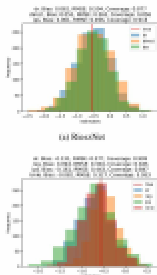

Figure: Bias, RMSE, and coverage for RieszNet and ForestRiesz on BHP semi-synthetic data with a complex DGP including linear and nonlinear confounders.

Ablation: Why Multitask and End-to-End Matter (RieszNet)

- We ablate multitasking and end-to-end on IHDP.
- Baseline RieszNet-DR: bias -0.044, RMSE 0.147, coverage 0.950.
- Separate NNs: DR bias -0.176, RMSE 0.411, coverage 0.880.
- No end-to-end: DR bias -0.051, RMSE 1.221, coverage 0.650.
- TMLE post-processing: DR bias -0.088, RMSE 0.182, coverage 0.950.
- Takeaway: multitasking and end-to-end yield the best accuracy and coverage; post-TMLE restores coverage with slightly higher bias.

Practical Guidance: Choosing RieszNet vs ForestRiesz

- We find RieszNet is flexible and end-to-end, excelling on average treatment effects with strong accuracy and coverage.
- We find ForestRiesz is structured and locally linear, shining on derivatives, especially with post-TMLE.
- We stabilize inference under extreme propensities by learning the Riesz representer directly.

Limitations and Failure Modes

- We require a black-box evaluation oracle for the functional.
- We recognize that double cross-fitting is incompatible with multitasking, creating trade-offs.
- We note finite-sample sensitivity to the choice of basis in ForestRiesz.
- We may face failures under severe misspecification of the functional oracle or extreme support violations with little overlap.
- Computational cost and tuning: training large neural networks and forests with cross-fitting increases runtime, and hyperparameter choices can affect finite-sample stability.

Conclusion: Turnkey Auto-DML for Linear Functionals

- We automate debiasing by learning Riesz representers with neural networks and forests.
- We deliver root-n accuracy and near-nominal coverage in practice.
- We outperform strong plug-in baselines on average treatment effects and average derivatives.

Future Work & Impact

- We plan to extend to richer functionals and structured treatments.
- We aim to automate basis selection and improve cross-fitting for multiple outcomes.
- We seek integration with policy gradients and reinforcement learning for uncertainty-aware optimization.

Questions & Discussion

- We appreciate your attention.
- We welcome questions and feedback.

Acknowledgments

- We thank collaborators, funding agencies, and open-source contributors.
- Code available at `https://github.com/victor5as/RieszLearning`